

## Lecture 21: Oct 10

Last time

- Internal midterm evaluation open

Today

- Moments

Moments

**Example** (Minimizing distance) The expected value of a random variable has another property, one that we can think of as relating to the interpretation of  $EX$  as a good guess at a value of  $X$ .

Suppose we measure the distance between a random variable  $X$  and a constant  $b$  by  $(X - b)^2$ . The closer  $b$  is to  $X$ , the smaller this quantity is. We can now determine the value of  $b$  that minimizes  $E[(X - b)^2]$  and, hence, will provide us with a good predictor of  $X$ . (Note that it does no good to look for a value of  $b$  that minimizes  $(X - b)^2$ , since the answer would depend on  $X$ , making it a useless predictor of  $X$ .)

We could proceed with the minimization of  $E(X - b)^2$  by using calculus, but there is a simpler method:

$$\begin{aligned} E(X - b)^2 &= E(X - EX + EX - b)^2 \\ &= E[(X - EX) + (EX - b)]^2 \\ &= E(X - EX)^2 + (EX - b)^2 + 2E[(X - EX)(EX - b)], \end{aligned}$$

where we have expanded the square. Note that  $E[(X - EX)(EX - b)] = (EX - b)E(X - EX) = 0$ , since  $EX - b$  is constant and comes out of the expectation,  $E(X - EX) = EX - EX = 0$ . This means

$$E(X - b)^2 = E(X - EX)^2 + (EX - b)^2.$$

Such that  $E(X - b)^2$  is minimized at  $b = EX$ . And  $E(X - EX)^2$  is actually the variance of  $X$  ( $Var X = E(X - EX)^2$ ).

The various moments of a distribution are an important class of expectations.

**Definition** For each integer  $n$ , the  $n$ th *moment* of  $X$  (or  $F_X(x)$ ),  $\mu'_n$ , is

$$\mu'_n = EX^n.$$

The  $n$ th *central moment* of  $X$ ,  $\mu_n$ , is

$$\mu_n = E(X - \mu)^n,$$

where  $\mu = \mu'_1 = EX$ .

Notes:

- $\mu'_0 = EX^0 = 1$
- $\mu'_1$  is the *mean*, usually denoted by  $\mu$ .
- $\mu_0 = E(X - \mu)^0 = 1$
- $\mu_1 = 0$
- $\mu_2 = E(X - EX)^2$  is the *variance*
- $\mu_3 = E(X - EX)^3$  is related to the *skewness*.
- $\mu_4 = E(X - EX)^4$  is related to the *kurtosis*.

**Definition** The *variance* of a random variable  $X$  is its second central moment,  $\text{Var}(X) = E[(X - EX)^2]$ . The positive square root of  $\text{Var}(X)$  is the *standard deviation* of  $X$ .

The variance gives a measure of the degree of spread of a distribution around its mean. Figure 20.1 shows a plot of two samples, one sample draws 100 numbers from a normal distribution with mean 0 and variance 1,  $N(0, 1)$ . The other sample draws 100 numbers from a normal distribution with mean 0 and variance 100,  $N(0, 100)$ .

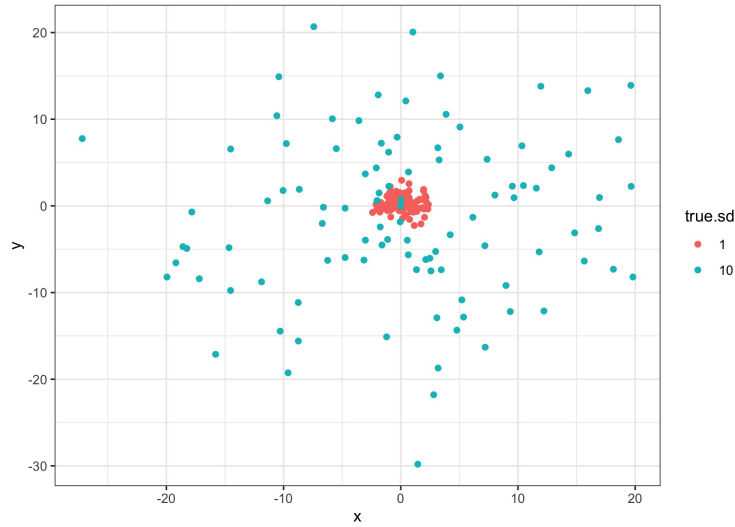


Figure 20.1: Figure 2.1.2. Two samples of 100 numbers drawn from  $N(0, 1)$  and  $N(0, 100)$ .

**Example** (Exponential variance) Let  $X$  have the exponential( $\lambda$ ) distribution. We can calculate the variance of  $X$  now.

*Solution:*

$$\begin{aligned} \text{Var}(X) &= E(X - \lambda)^2 \\ &= \int_0^{\infty} (x - \lambda)^2 \frac{1}{\lambda} e^{-x/\lambda} dx \end{aligned}$$

**Theorem** If  $X$  is a random variable with finite variance, then for any constants  $a$  and  $b$ ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

*Proof:*

From the definition, we have

$$\begin{aligned} \text{Var}(aX + b) &= E[(aX + b) - E(aX + b)]^2 \\ &= E(aX - aEX)^2 \\ &= a^2 E(X - EX)^2 \\ &= a^2 \text{Var}(X). \end{aligned}$$

It is sometimes to use an alternative formula for the variance, given by

$$\text{Var}(X) = E(X^2) - (EX)^2,$$

which is easily established by

$$\begin{aligned} \text{Var}(X) &= E(X - EX)^2 = E[X^2 - 2XEX + (EX)^2] \\ &= EX^2 - 2(EX)^2 + (EX)^2 \\ &= EX^2 - (EX)^2. \end{aligned}$$

**Example** (Binomial variance) Let  $X \sim \text{Binomial}(n, p)$ , that is ,

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

What is the variance of  $X$ ?

*Solutions:*

**Definition** Let  $X$  be a random variable with cdf  $F_X$ . The *moment generating function (mgf)* of  $X$  (or  $F_X$ ), denoted by  $M_X(t)$ , is

$$M_X(t) = Ee^{tX},$$

provided that the expectation exists for  $t$  in some neighborhood of 0. That is, there is an  $h > 0$  such that, for all  $t$  in  $-h < t < h$ ,  $Ee^{tX}$  exists. If the expectation does not exist in a neighborhood of 0, we say that the moment generating function does not exist.

More explicitly, we can write the mgf of  $X$  as

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx, \quad \text{if } X \text{ is continuous,}$$

or

$$M_X(t) = \sum_x e^{tx} \Pr(X = x), \quad \text{if } X \text{ is discrete.}$$

It is easy to see how the mgf generates moments as in the following theorem.

**Theorem** If  $X$  has mgf  $M_X(t)$ , then

$$EX^n = M_X^{(n)}(0),$$

where we define

$$M_X^{(n)}(0) = \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0}.$$

That is, the  $n^{th}$  moment is equal to the  $n^{th}$  derivative of  $M_X(t)$  evaluated at  $t = 0$ .

*Proof:*

**Example** (Binomial mgf) Let  $X \sim \text{Binomial}(n, p)$ , then its mgf is

$$\begin{aligned} M_X(t) &= \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x} \\ &= [pe^t + (1-p)]^n. \end{aligned}$$

**Theorem** Let  $F_X(x)$  and  $F_Y(y)$  be two cdfs all of whose moments exist.

1. If  $X$  and  $Y$  have **bounded support**, then  $F_X(u) = F_Y(u)$  for all  $u$  if and only if  $EX^r = EY^r$  for all integers  $r = 0, 1, 2, \dots$ .
2. If the moment generating functions exist and  $M_X(t) = M_Y(t)$  for all  $t$  in some neighborhood of 0, then  $F_X(u) = F_Y(u)$  for all  $u$ .

**Theorem** (Convergence of mgfs) Suppose  $\{X_i, i = 1, 2, \dots\}$  is a sequence of random variables, each with mgf  $M_{X_i}(t)$ . Furthermore, suppose that

$$\lim_{i \rightarrow \infty} M_{X_i}(t) = M_X(t), \quad \text{for all } t \text{ in a neighborhood of } 0,$$

and  $M_X(t)$  is an mgf. Then there is a unique cdf  $F_X$  whose moments are determined by  $M_X(t)$  and, for all  $x$  where  $F_X(x)$  is continuous, we have

$$\lim_{i \rightarrow \infty} F_{X_i}(x) = F_X(x).$$

That is, *convergence*, for  $|t| < h$ , of mgfs to an mgf implies *convergence* of cdfs.

**Poisson approximation** One approximation that is usually taught in elementary statistics courses is that binomial probabilities can be approximated by Poisson probabilities. It is taught that the Poisson approximation is valid “when  $n$  is large and  $np$  is small”, and rules of thumb are sometimes given.

The *Poisson*( $\lambda$ ) pmf is given by

$$\Pr(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots,$$

where  $\lambda$  is a positive constant. The approximation states that if  $X \sim \text{Binomial}(n, p)$  and  $Y \sim \text{Poisson}(\lambda)$ , with  $\lambda = np$ , then

$$\Pr(X = x) \approx \Pr(Y = x)$$

for large  $n$  and small  $np$ . We now show that the mgf converge, lending credence to this approximation. Recall that

$$M_X(t) = [pe^t + (1 - p)]^n.$$

For the *Poisson*( $\lambda$ ) distribution, we can calculate (HW4, exercise 2.33)

$$M_Y(t) = e^{\lambda(e^t - 1)},$$

and if we define  $p = \lambda/n$ , then  $M_X(t) = [1 + (e^t - 1)\lambda/n]^n$  such that  $M_X(t) \rightarrow M_Y(t)$  as  $n \rightarrow \infty$ .

**Theorem** For any constant  $a$  and  $b$ , the mgf of the random variable  $aX + b$  is given by

$$M_{aX+b} = e^{bt} M_X(at).$$

*Proof:*

## Common Discrete Distribution

Why parametric models?

- *Parametric models* or *distribution families* have a specific form but can change according to a fixed number of parameters.
- The objective is to model a population. Parametric models are often appropriate in common situations with similar mechanisms.
- Parametric models have many known and useful properties and are easy to work with. When fitting a population, only a few parameters need to be estimated: *parametric inference*.
- Sometimes one does not want to make parametric assumptions and would rather work with non-parametric models. But non-parametric models can be infinite dimensional.
- In this course, we emphasize parametric models.

**Discrete uniform**  $X$  has the discrete uniform( $1, N$ ) distribution if  $X$  is equally likely to be one of  $\{1, 2, \dots, N\}$ .

- Sample space:  $\{1, 2, \dots, N\}$

- pmf:

$$f_X(x) = \frac{1}{N}, \quad x = 1, 2, \dots, N$$

- cdf:

$$F_X(x) = \Pr(X \leq x) = \begin{cases} 0 & x < 0 \\ \lfloor x \rfloor / N & 0 \leq x < N \\ 1 & N \leq x \end{cases}$$

- moments:

$$EX = \frac{N+1}{2}$$